

A Prym–Petri theorem for étale μ_3 -covers at the depth-one stratum

Samuel Lavery
sam@sk-labs.com

July 19, 2026 — v0.1

Abstract

Let $\pi : \tilde{C} \rightarrow C$ be a connected étale $\mathbb{Z}/3$ -cover of a smooth curve of genus g , defined by a 3-torsion class $\varepsilon \neq 0$. For $L \in \mathrm{Nm}^{-1}(\omega_C)$ the Prym–Petri map is the anti-invariant projection of the Petri multiplication; its kernel always contains a canonical trace line T_L . We prove: at the depth-one stratum $h^0(L) = 1$, the reduced map is injective for every L and every $\varepsilon \neq 0$, provided C avoids an explicit Brill–Noether-type locus — unconditionally at $g \leq 5$ (in particular $g = 4$, the genus of sixfold Pryms), and at general g modulo one remaining window. The method is a complete reduction to base-curve divisor geometry: the kernel equals h^0 of the *collision excess* of the section divisor, and failure occurs exactly when that excess moves. The generality hypothesis is necessary: on any curve with a vanishing thetanull the failure is constructed explicitly, and it occurs inside the depth-one stratum. To our knowledge (checked against the literature at source, including the full citation graph of Welters’ theorem) no Prym–Petri statement for cyclic covers of degree ≥ 3 existed; all prior extensions of [1] are for double covers. This paper does not assume or prove RH/GRH.

1 Statement

Throughout, C is a smooth projective curve of genus $g \geq 2$ over an algebraically closed field of characteristic $\neq 3$, $\varepsilon \in \mathrm{Pic}^0(C)[3]$ is nonzero, $\pi : \tilde{C} \rightarrow C$ the associated connected étale cyclic cover with deck generator σ , and Nm the norm. Then $\omega_{\tilde{C}} = \pi^*\omega_C$ and $\tilde{g} = 3g - 2$. For $L \in \mathrm{Nm}^{-1}(\omega_C)$ one has $\deg L = 2g - 2$, $\chi(L) = 1 - g$, and $L \otimes \sigma^*L \otimes \sigma^{2*}L \cong \pi^*\omega_C$.

The Petri multiplication $\alpha_L : H^0(L) \otimes H^0(\omega_{\tilde{C}}L^{-1}) \rightarrow H^0(\omega_{\tilde{C}})$ composed with the projection pr onto $H^0(\omega_C\varepsilon) \oplus H^0(\omega_C\varepsilon^2)$ (the anti-invariant part) is the μ_3 -Prym–Petri map β_L . For $0 \neq s \in H^0(L)$ the element $s \otimes (\sigma^*s \cdot \sigma^{2*}s)$ maps to the σ -invariant norm section; at $h^0(L) = 1$ these span the *trace line* $T_L \subseteq \ker \beta_L$.

Theorem 1.1 (μ_3 -Prym–Petri, depth one). *Suppose C lies outside the proper closed locus of Lemma 3.1 below. Then for every $\varepsilon \neq 0$ and every $L \in \mathrm{Nm}^{-1}(\omega_C)$ with $h^0(L) = 1$,*

$$\ker \beta_L = T_L.$$

This holds unconditionally (given the cited classical inputs) for $g \leq 5$; for $g \geq 6$ it holds outside the window described in §3.

Theorem 1.2 (Sharpness). *Let C carry a vanishing thetanull: $\omega_C = 2N$ with N a pencil. Then for every $\varepsilon \neq 0$ the conclusion of Theorem 1.1 fails: for any reduced $E_0 \in |N|$, distributing two*

of the three preimages of each point of E_0 produces W on $\tilde{C}$ with $L = \mathcal{O}(W) \in \text{Nm}^{-1}(\omega_C)$ and $\dim\{t : \alpha_L(s \otimes t) \text{ } \sigma\text{-invariant}\} = h^0(N) = 2 > 1$. On the verification curve (§5) these L have $h^0(L) = 1$, so the failure occurs inside the depth-one stratum.

Remark 1.3. The statement of Theorem 1.1 has no analogue at depth one for $\mathbb{Z}/2$: there $\chi(L) = 0$, so $h^0(\omega_{\tilde{C}}L^{-1}) = 1$ and the reduced domain is zero — Welters’ theorem [1] begins at $h^0 \geq 2$. The depth-one μ_3 statement, its reduction (Lemma 2.1), and Theorem 1.2 appear to be new; every extension of [1] we located (pointed [5], two-pointed [6], ramified double [7]) concerns degree-2 covers, and the only Brill–Noether study of higher cyclic covers [8] contains no Petri-type statement.

2 The reduction to collision geometry

For an effective divisor W on $\tilde{C}$ define the *max-descent* $\bar{D} = \sum_x \max_{y \in \pi^{-1}(x)} \text{mult}_y(W) \cdot x$ on C and the *collision excess* $E = \pi_*W - \bar{D} \geq 0$, supported in $\text{supp } \bar{D}$.

Lemma 2.1 (Reduction). *Let $h^0(L) = 1$ with section s , $W = \text{div}(s)$. Then*

$$\dim \ker(\text{pr} \circ \alpha_L) = h^0(\omega_C - \bar{D}), \quad \omega_C - \bar{D} \sim E,$$

so $\ker \beta_L = T_L$ if and only if $h^0(\mathcal{O}_C(E)) = 1$, i.e. iff the collision excess does not move.

Proof. Every domain element is $s \otimes t$. Now $\text{pr}(\alpha(s \otimes t)) = 0$ iff $s \cdot t$ lies in the σ -invariant part of $H^0(\pi^*\omega_C)$, which is $\pi^*H^0(\omega_C)$ (étale, $\text{char} \neq 3$, averaging). And $s \cdot t = \pi^*\eta$ iff $t = \pi^*\eta/s$ is regular iff $\pi^*\text{div}(\eta) \geq W$ iff $\text{div}(\eta) \geq \bar{D}$, the last step pointwise on fibers. Thus $\{t\} \cong H^0(\omega_C - \bar{D})$, which contains the norm differential $\eta_s = \text{Nm}(s)$ with $\text{div}(\eta_s) = \pi_*W$; its tensor is exactly T_L . Finally $\pi_*W \in |\omega_C|$ gives $\omega_C - \bar{D} \sim \pi_*W - \bar{D} = E$. $\square$

Corollary 2.2. *Collision-free sections ($E = 0$) never lie in a failing kernel.*

Lemma 2.3 (Collision arithmetic). *Write fiber multiplicities (a, b, c) with $a = \text{mult}_x \bar{D}$ maximal and $k_x = \text{mult}_x E = b + c$. Then $b, c \leq a$ forces $a \geq \lceil k_x/2 \rceil$, so the canonical divisor $K = \pi_*W$ satisfies $K \geq E + \lceil E/2 \rceil$, where $\lceil E/2 \rceil = \sum \lceil k_x/2 \rceil x$. Conversely any moving E and canonical $K \geq E + \lceil E/2 \rceil$ assemble to a failing candidate. (For μ_n : $K \geq E + \lceil E/(n-1) \rceil$.)*

Failure at depth one is therefore *equivalent* to: some moving effective E admits a canonical divisor containing its half-thickening. All μ_3 -structure is now consumed; what remains is base-curve Brill–Noether theory, uniform in ε .

3 The base-curve exclusion

Lemma 3.1 (Exclusion). *For C general in $\mathcal{M}_g$, no moving effective E has $h^0(\omega_C - E - \lceil E/2 \rceil) > 0$. Status: proven for $g \leq 5$; for $g \geq 6$ proven outside the window $\frac{g+2}{2} \leq e \leq \frac{2g}{3}$ (with $e = \deg E$; at $g = 6$ only the single boundary point $e = 4$ remains).*

The proof combines three ingredients.

(Z1) *Zone arithmetic.* Write $E = \sum k_i x_i$ over r points, $\delta = e - r$, $c = \sum \lceil k_i/2 \rceil$. Pointwise $\lceil k/2 \rceil + (k-1) \geq k$, so $c + \delta \geq e$. In the automatic-speciality zone $e + c \leq g - 1$, membership of a profile- δ divisor in a pencil costs $\delta \leq \rho(g, 1, e) + 1 = 2e - g - 1$ (valid for *all* pencils on a general

curve by [2, Thm. 1.1] applied at full degree), whence $c + \delta \leq (g - 1 - e) + (2e - g - 1) = e - 2 < e$: contradiction. The zone is empty for all profiles and all g .

(Z2) *Virtual dimension.* Outside the zone the bad locus has virtual dimension $r - c - 2 \leq -2$, since $c \geq r$.

(Comparison kill, $\delta = 0$.) If some pencil N of degree e has $s = h^0(\omega - 2N) \geq 1$, then every reduced $E \in |N|$ satisfies $K \geq 2E$ for some canonical K (a class-level statement), so the de Jonquière locus of the canonical series with profile (2^e) and $f = |\mu| - r$ contains a family of dimension $\dim W_e^1 + 1 = 2e - g - 1$. By [2, Thm. 1.1] — whose hypotheses we verified at source: closed determinantal locus, no distinctness requirement, boundary f allowed, every series covered — each component has dimension exactly $g - e - 1$. Hence $2e - g - 1 \leq g - e - 1$, i.e. $e \leq 2g/3$. Since moving E forces $e \geq (g + 2)/2 > 2g/3$ for $g \leq 5$, the $\delta = 0$ case is complete there; at $g = 4$, every $e \geq 3$ exceeds $8/3$. (At $g = 6$, $e = 4$ the comparison gives equality and does not kill; this boundary point, of Brill–Noether number $\rho = 0$, has exactly the residual-pair shape of [4, Thm. 1.1], and joins the window. This correction — from an earlier “ $g \leq 6$ ” claim — was caught during the Lean formalization; see §5.)

At $g = 4$ the remaining $\delta = 1$ profiles reduce to finitely many ramification coincidences of the trigonal pencils, each a proper closed condition; the $\delta = 0$ statement at $e = g - 1 = 3$ is classically equivalent to the absence of a vanishing theta null. A caution for future use: the non-existence theorem of [3, Thm. 1.5] requires the series *general* in G_d^r — the quotation of it in [2] omits this hypothesis, and composed series such as $|2N| \supseteq 2|N|$ show the unrestricted reading is false. It is therefore not usable against special pencils, and we avoid it entirely.

Proof of Theorem 1.2. $\omega = 2N$ makes $K = 2E_0$ canonical for every $E_0 \in |N|$; the construction of Lemma 2.3 with $(a, b, c) = (1, 1, 0)$ over each point gives W with $\overline{D} = E_0$, and Lemma 2.1’s count is $h^0(\omega - E_0) = h^0(N) = 2$. $\square$

4 The depth-one threefold and the μ_3 parity

Let $Y = \mathrm{Nm}^{-1}(\omega_C)$ (a torsor of dimension $2g - 2$, tangent space $H^1(\varepsilon) \oplus H^1(\varepsilon^2)$) and $V^1 = \{L \in Y : h^0(L) \geq 1\}$. Theorem 1.1 has a geometric equivalent.

Theorem 4.1 (Tangent lemma). *At a depth-one $L \in V^1$: $T_L V^1 = \mathrm{ann}(\mathrm{Im} \beta_L)$, and $\dim_L V^1 = g - 1$ always. Hence $\ker \beta_L = T_L$ if and only if V^1 is smooth of dimension $g - 1$ at L ; Theorem 1.1 says V^1 is smooth along its whole depth-one locus for (C, ε) general, and Theorem 1.2 exhibits, on theta null curves, a singular curve inside V^1 swept by the trigonal-fiber constructions.*

Proof. First-order deformations preserving the section are cut by $v \cup s$; Serre-dually the transpose of $\cup s$ is the s -slice of α_L , and restriction to $T_L Y$ replaces α by β . The dimension statement: the assignment $\eta \in |\omega_C| \rightsquigarrow$ preimage splittings W of $\mathrm{div}(\eta) \rightsquigarrow L = \mathcal{O}(W)$ is a surjection onto V^1 , finite over $|\omega_C| = \mathbb{P}^{g-1}$, with fiber $|L|$ over a depth-one L . $\square$

Theorem 4.2 (μ_3 parity). *V^1 has exactly three irreducible components, indexed by the total sheet-shift $\Sigma \in \mathbb{Z}/3$ of the splitting W relative to the monodromy: as η loops in $|\omega_C|$, the cycle traced by $\mathrm{div}(\eta)$ bounds the swept 2-chain, hence is nullhomologous and carries zero ε -holonomy; the monodromy of the 3^{2g-2} splittings is therefore the index-3 subgroup $\{\sum a_i = 0\} \rtimes S_{2g-2}$ of the wreath product, with three orbits. This is the $\mathbb{Z}/3$ -analogue of Mumford’s parity for classical Pryms. Consequently, for a divisor with closed points of degrees $d_1, \dots, d_r$ over $\mathbb{F}_q$, the rational*

splittings distribute over the components uniformly if some $d_i \not\equiv 0 \pmod{3}$, and lie in a single component if all $d_i \equiv 0 \pmod{3}$ (Lean: `MuThreePrymParity.uniformity/concentration`).

On the verification curve the parity law is confirmed exactly: the $\mathbb{F}_{49}$ census gives $341552/49^3 = 2.903$ (three components, deficit $0.68q^{5/2}$), and all 48 clean all-split $\mathbb{F}_7$ divisors obey the distribution law — degree profiles (1, 2, 3), (1, 5), (2, 4) split (9, 9, 9), (3, 3, 3), (3, 3, 3); profile (6) concentrates (3, 0, 0), twenty cases. The Gauss map of V^1 at a generic depth-one point has full rank 3 (second-fundamental-form blocks of ranks (2, 3, 3)): the components are nondegenerate, theta-like threefolds.

5 Verification register

All claims above are tiered: Lemmas 2.1, 2.3 and Theorem 1.2 are complete proofs; Lemma 3.1 is proven in the stated range on the cited inputs; Theorem 1.1 inherits exactly that range. The open window at $g \geq 6$ is a genuine gap, named above, localized in the Farkas–Ungureanu framework.

Machine verification (exact arithmetic over $\mathbb{F}_{7^6}$ on the tower $E_0: y^2 + 2xy + y = x^3$, $C: u^3 + xu + (y+1) = 0$ of genus 4, $\tilde{C}: t^3 = y$ of genus 10): `petri_t1.sage/petri_t1b.sage` (eigenspace gates 4/3/3; depth-one battery: $\ker \beta = T_L$ exactly, 14/14); `chain_occupancy.sage` (twisted limit canonicals on the μ_3 -chain: flat, tail-nested, 36+ configurations); `thetanull.check.sage` (the tower curve is a cone: quadric rank 3, $\omega = 2N$ with N the u -map; predicted kernel jump to $h^0(N) = 2$ on u -fiber triples 5/5, control triples kernel 1, 5/5); `hw.check.sage` ($h^0(\mathcal{O}(W)) = 1$ in 10/12 constructed configurations: depth-one failures literal). The 14/14 battery is *explained* by Corollary 2.2: random sections are collision-free.

Further instruments for §4: `w2.specimen.sage` (the bielliptic non-thetanull tower, quadric rank certified in closed form), `w2.battery.sage` (100 configs, all 54 depth-one ones smooth points of V^1), `w3.count.sage/w3.count2.sage` (the $\mathbb{F}_7/\mathbb{F}_{49}$ censuses), `w3.gauss.sage` (Gauss rank), `w3.parity.sage` (the 48/48 distribution law). The falsification record of §4: a first-draft irreducibility claim for V^1 was refuted by the $\mathbb{F}_{49}$ census within the hour; the corrected monodromy constraint (the nullhomologous divisor-motion cycle) *is* the parity mechanism.

Lean (`RequestProject/MuThreePrymPetri.lean` and `MuThreePrymParity.lean`, build clean, axiom footprint `{propext, Classical.choice, Quot.sound}`): `FiberTriple.canonical_ge_three_half` (Lemma 2.3); `Profile.zone_empty (Z1)`; `Profile.virtual_dim_le_neg_two (Z2)`; `dimension.comparison, g4_kill, delta0_excluded_g4` (the comparison kill); and `g6_boundary_not_killed`, the honesty marker recording that $g = 6$, $e = 4$ is *not* covered.

Provenance. The novelty claim was checked before being made: five independent sweeps including the complete citation graph of [1] (80 works) found no cyclic-degree- ≥ 3 Prym–Petri statement; residual risk (non-arXiv venues, non-citing papers) is acknowledged. One correction is part of this record: an initial “ $g \leq 6$ ” coverage claim for the comparison kill was corrected to “ $g \leq 5$ plus boundary” when the Lean formalization exposed the equality case.

References

- [1] G. E. Welters, *A theorem of Gieseker–Petri type for Prym varieties*, Ann. Sci. Éc. Norm. Supér. (4) **18** (1985), 671–683.
- [2] G. Farkas, *Generalized De Jonquières divisors on generic curves*, arXiv:2210.07843.

- [3] M. Ungureanu, *Dimension theory and degenerations of de Jonquières divisors*, arXiv:1612.03141.
- [4] M. Ungureanu, *Geometry of intersections of some secant varieties to algebraic curves*, arXiv:1810.05461.
- [5] N. Tarasca, *A pointed Prym–Petri theorem*, arXiv:2202.05284.
- [6] M. Jeon, *Two-pointed Prym–Brill–Noether loci and coupled Prym–Petri theorem*, arXiv:2309.02642.
- [7] A. Bud, *Prym–Brill–Noether theory for ramified double covers*, arXiv:2411.00716.
- [8] I. Schwarz, *Brill–Noether theory for cyclic covers*, arXiv:1603.05084.